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Genome wide identification of MATE and ALMT gene family in lentil (*Lens culinaris* Medikus) and expression profiling under Al stress condition

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Abstract

Background The membrane transporters viz. multidrug and toxic compound extrusion (MATE) and aluminum-activated malate transporter (ALMT) are associated with aluminum (Al) tolerance by accelerating secretion of organic acids, which can influence nutrient availability and stress response. However, such transporter families have not yet been reported in lentil under Al stress condition.

Method and results In this study, 90 MATE and 14 ALMT genes were identified and clustered into four (MATE) and five (ALMT) subfamilies/clades with smaller subgroups. All the MATE and ALMT genes were unevenly dispersed across lentil chromosomes. Duplication analysis suggested that LcMATE gene family has expanded primarily through tandem duplication event. Collinearity of lentil with soybean suggested a close relationship between the MATE genes. The MATE promoter regions harboured many stress responsive as well as Al resistance transcription factor 1 related cis-regulatory elements. Predicted 3D (three-dimensional) structure and molecular docking revealed that 5 LcMATE proteins could bind citrate and contain amino acids related to its secretion *via* citrate exuding motif and other neighbouring sites. Expression analyses of LcMATE and LcALMT genes were performed using quantitative real-time polymerase chain reaction (qRT-PCR). Six genes namely, LcM1, LcM42, LcM46, LcM47, LcALMT8 and LcALMT14 responded to Al stress with varying levels of expression patterns at different time points (3, 6, 12 and 24 h).

Conclusion Our findings offer thorough details on the MATE and ALMT transporters in lentils and will aid in valuable understanding for future functional studies of these transporters in generating Al tolerant cultivars.

Keywords ALMT, Aluminum, Expression profiling, Gene families, Lentil, MATE, Molecular docking, Transporter

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Introduction

Aluminium toxicity is one of the major threats that hinder crop productivity under acidic soils. When pH of the soil is <5.5, Al solubilizes and is available in the form of most toxic trivalent cation Al^{3+} [30]. Cationic Al^{3+} mostly prevents root elongation, limiting food and water uptake and significantly lowers seed output. Plant species have developed various mechanisms to diminish Al toxicity under acidic soils [47]. These mechanisms can be classified into two groups: (i) Al exclusion mechanism that restricts Al from the root by secreting organic acids to chelate Al cations and (ii) Al tolerance mechanisms in which Al that entered the root and shoot is detoxified through sequestration into vacuoles [31]. In plants, membrane transporters like multidrug and toxic compound extrusion (MATE) and aluminium activated malate transporter (ALMT) are involved in conferring Al tolerance through the release of organic acids from the root tips [41].

In the plant kingdom, MATE gene family is known to play important and varied roles under biotic and abiotic stress conditions [28]. The MATE genes have been characterized and identified in various plant species under aluminium stress conditions including maize [50, 88], rice bean [43], *Arabidopsis* [75], rice [84], soybean [48], pigeonpea [7], barley [17], and rye [16]. These MATE genes are known to confer tolerance against Al toxicity in plants. They are localised to plasma membranes at the tips of the roots. For example, in case of barley and sorghum, the MATE gene encoded a citrate flux transporter which contributes significantly to Al resistance [17, 49]. Furthermore, their homologs were then recognised as *AtMATE1* in *Arabidopsis* [36], *ZmMATE1* in maize [50], *OsFRDL4* in rice [83] and *VuMATE1* in rice bean [43]. The citrate transporter *CcMATE35* in pigeon pea was identified to be involved in Al stress response [7]. Map based cloning was utilized to identify first MATE transporter from *Sorghum bicolor* (*SbAMTE*) [49] and *Hordeum vulgare* (*HvAACT1*) [17] which are involved in Al detoxification. The secretion of citrate is facilitated by *HvAACT1*, which encodes for an enzyme in the multidrug and toxic compound extrusion (MATE) family [86]. MATE from cabbage (*Brassica oleracea*, *BoMATE*) enhanced Al tolerance in *A. thaliana* via citrate efflux in presence of Al^{3+} [81]. In addition, MATE genes also participate in secondary metabolite transport, disease resistance and plant hormone transport [64].

Another gene family, aluminium activated malate transporter (ALMT) is also involved in Al tolerance, pH regulation, stomatal movement, and mineral nutrition [10, 22, 38, 39]. Various genes encoding ALMT involved in imparting tolerance against Al toxicity have been found in many plant species, such as *Triticum aestivum*

(*TaALMT1*) [65], *Brassica napus* (*BnALMT1*, *BnALMT2*) [38], *Arabidopsis thaliana* (*AtALMT1*) [25], *Secale cereale* (*ScALMT1*) [16], *Glycine max* (*GmALMT1*) [37], and *Hordeum vulgare* (*HvALMT1*) [22]. Studies have also shown that ALMT transporters are also involved in other physiological processes including cellular signaling, fruit flavor regulation, cell expansion, nutrients acquisition and GABA (γ -aminobutyric acid) signalling [67].

Lentil is one of the cool season food legumes and plays a significant role as staple food for humans and fodder for animals [12]. However, its growth is seriously affected by Al stress condition [70]. Aluminium resistance in lentil was first reported by Singh et al. [70] and they also found that secretion of organic acids was significantly higher in Al-tolerant than Al-sensitive genotypes of lentil. Transcriptomic study in root tips of lentil indicated that MATE and ALMT genes were up-regulated in response to Al stress [68]. Several studies have been conducted to identify MATE and ALMT transporters in response to Al toxicity in various crop species viz., mungbean, pigeon pea, wheat, maize, soybean, and rice [7, 11, 40, 50, 71, 84]. Nonetheless, as far as we know, no efforts have been undertaken to investigate the genome-wide studies of ALMT and MATE transporters in lentil exposed to Al stress. Therefore, the current investigation was conducted in order to (i) survey the entire genome of lentil for potential MATE and ALMT transporters as well as their chromosomal location, phylogenetic connection, protein motifs and structures, (ii) analyse their expression dynamics between contrasting genotypes (Al tolerant and Al-sensitive) at the root tips.

Methodology

Screening and identification of MATE and ALMT transporters in lentil

The protein sequences of a total of 56 MATE transporters (PF01554) and 14 ALMT transporters (PF11744) identified in *Arabidopsis* were gathered from Phytozome. Plausible MATE and ALMT protein sequences of lentil were obtained using these sequences as seed sequences for BLASTP searches against the protein dataset of *Lens culinaris* CDC Redberry Genome 2.0 (E-value = $\leq 1e-7$). These putative MATE and ALMT sequences were refined by identification of conserved MATE (Pfam: PF01554) and ALMT (PF11744) domains using the Pfam database [52]. Data files with the coding, genomic and protein sequences for the identified lentil MATE and ALMT genes were obtained from Phytozome (Files S1 and S2). The biophysical properties including the theoretical isoelectric point and molecular weight were calculated using EXPASY database (<http://expasy.org/>) [21]. The subcellular localization and number of transmembrane helices were predicted

by WOLFP SORT (<https://wolfsort.hgc.jp/>) [26] and DeepTMHMM (<https://dtu.biolib.com/DeepTMHMM>) [23], respectively.

Chromosomal location, gene duplication and collinearity analysis

The chromosomal locations of each LcMATE and LcALMT genes were determined from the Knowpulse web-resource (<https://knowpulse.usask.ca/lentil-genome>) [63]. A pictorial form of the chromosomal location was given by MapChart [77]. Duplications were identified by multiple collinearity scan toolkit (MCScan) [79] with default settings, wherein BLASTP was used to search for potential anchors ($E < 1e-5$, top 5 matches). Tandem duplication was defined as homologs on a chromosome less than 10 gene loci apart and more than 50% protein similarity. Duplicate genes were classified into dispersed, proximal, tandem and segmental duplications labelled as 1, 2, 3 and 4, respectively; while singletons were labelled as 0. The McScanX and TBtools (version 2.008) were used for collinearity and Ka/Ks analysis. The ratio of non-synonymous (Ka) substitutions to synonymous (Ks) substitutions for each collinear MATE gene pair was used to assess selection pressure. Value of Ka/Ks ratio less than, more than and equal to 1 were denoted as purifying, positive and neutral selection pressure, respectively. The divergence time (T) for each collinear gene pair was estimated using the following equation: $T = Ks / (2 \times 6.5 \times 10^{-9}) \times 10^{-6}$ million years ago (MYA) [44].

Phylogenetic and structural analyses of MATE and ALMT transporters in lentil

The protein sequences of previously reported MATE transporters (33) and ALMT transporters (29) from other plant species retrieved from NCBI were used for phylogenetic analysis (Table S1 and File S3). The identified protein sequences of 90 lentil MATEs and 14 lentil ALMTs (Files S1 and S2) along with their previously reported homologs were separately used for multiple sequence alignments by ClustalW followed by construction of the unrooted phylogenetic trees by MEGA X [33] using the maximum likelihood algorithm with 1000 bootstraps. Structure of identified genes were analysed using the Gene Structure Display Server (GSDS) with default settings [27]. Motifs in MATE and ALMT proteins were identified using MEME (multiple EM for motif-elicitation) tool (<http://meme-suite.org/>) [1] with default settings. For MATE and ALMT, the maximum number of motifs were twelve and six, respectively.

Prediction of the cis-acting regulatory elements in the LcMATE genes

The recently added lentil genome on Phytozome provided the 2000 bp upstream promoter region with respect to the START codon of the identified LcMATE genes. The cis-acting elements were identified in the promoter region *via* PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) [35].

Three-dimensional (3D) structure prediction, molecular docking and binding site analyses of LcMATE

The online website PHYRE2 (<http://www.sbg.bio.ic.ac.uk/phyre2/html/page.cgi?id=index>) was used to analyse the accuracy of alignment and the 3D structure of LcMATE proteins grouped along with previously characterised MATE proteins known for their role in aluminum tolerance [29]. Molecular docking was performed with CB-Dock (<http://cao.labshare.cn/cb-dock/>) [45], to understand the interactions between citrate and selected LcMATE transporters. Multiple Sequence Alignment (MSA) was used to identify the Citrate Exuding Motif (CEM). Sequence similarity was also conducted for the LcMATEs grouped under the subgroup 3 of subfamily 2 using UNIPROT (<https://www.uniprot.org/align>) [5]. HMMTOP (<http://www.enzim.hu/hmmtop/index.php>) was used to predict the transmembrane helices and topology of LcM1, LcM42, LcM46, LcM47 and LcM74 [74].

Protein-protein interaction of LcMATE transporters

Protein-Protein Interactions (PPI) for the five LcMATE transporters grouped under subgroup 3 of subfamily 2 (S2-3) was analyzed using STRING v12.0 (<https://string-db.org/>) [51]. The PPI networks were created by integrating active interactions with an interaction score of > 0.3 from sources like text mining, gene fusion, co-expression and databases.

Plant material, growth conditions and aluminium stress treatment

The Al-tolerant (L-4602) and Al-sensitive (BM4) genotypes of *Lens culinaris* Medikus were phenotyped for their aluminium tolerance based on morpho-physiological traits under hydroponic and field conditions [70]. L-4602 was developed at Indian Council of Agricultural Research (ICAR)-Indian Agricultural Research Institute (IARI, New Delhi, India), while BM-4 was developed at the International Centre for Agricultural Research in the Dry Areas, Syria [18, 70]. These were evaluated in hydroponics at National Phytotron Facility, ICAR-IARI, New Delhi, India. The growth chamber's settings were set to maintain a temperature of 22/18°C (± 2 °C) day and night, along with a 45% of relative humidity. Low ionic

hydroponic medium containing micro and macronutrient as described by Singh et al. [69] was used for growth at seedling stage for qRT-PCR experiment. After seven days, uniform seedlings were chosen and subjected to two sets of conditions: control (0 μM $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$, pH 4.8) and Al stress (148 μM $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$, pH 4.8). Root samples from both the control and Al treated plants were taken at various intervals (0, 3, 6, 12 and 24 h). The collection was followed by freezing the root tips using liquid nitrogen and then they were kept at -80°C for further experiments. Three biological replicates, each including 12 seedlings under treatment and control conditions, were used in the entirely randomized design of the experiment.

RNA extraction and qRT-PCR analysis

Promega's SV total RNA isolation kit was used to isolate total RNA from <1 cm root tips of both the genotypes under Al treated and control conditions at different time points, following the manufacturer's protocol. Then, using a Revert Aid First Strand cDNA synthesis kit (Thermo Scientific, USA), first strand of cDNA was generated from 2 μg of isolated RNA. LcM1, LcM42, LcM46, LcM47, LcALMT8 and LcALMT14 were selected for differential gene expression analysis as these were probable homologs of previously characterized genes involved in Al detoxification *via* organic acid exudation.

The Real-time PCR primer and probe design tool (<https://www.genscript.com/tools/real-time-pcr-Taqman-primer-design-tool>) was used to create primers specific to the genes (Table S2). The qRT-PCR analysis was performed using the Insta Q96 Plus Real Time Machine PCR system (HiMedia, India). The 20 μL qRT-PCR mixes contained 10 μL of TB Green Premix Ex Taq (Tli RnaseH Plus), 0.8 μL of each forward and reverse primer, 0.4 μL of ROX Dye, and 1 μL of cDNA. The qRT-PCR temperature profile comprised of 40 cycles of 15 s at 95°C , 30 s at 60°C and 10 s at 72°C performed after 5 min at 95°C for initial denaturation. In order to guarantee the integrity and accuracy of the PCR products, the melting stage was conducted under default conditions. As an internal standard β -tubulin gene [13] was used to calculate the relative fold changes based on the comparative Ct method. The expression levels were calculated using the $2^{-\Delta\Delta\text{CT}}$ method [66].

Statistical analyses

All the data were analysed using SPSS program (version 24) [15]. The statistical significance of the difference was assessed at the 0.05 and 0.0001% level of significance using the student's t-test along with post-tests.

Results

Identification of MATE and ALMT genes in lentil

A total of 90 gene loci coding for MATE transporters were found in the reference genome of lentil. The MATE transporters were identified and designated LcM1 to LcM90 in accordance to their chromosomal locations. The length and molecular weight of lentil MATE proteins had a broader range varying from 129 to 1167 amino acids, and 13.66 to 123.9 kDa, respectively. Theoretical pI values were between 4.92 and 9.58. There were several transmembrane helices (TMHs) in the LcMATEs ranging from 1 to 12 (Table S3).

The prediction of sub-localization revealed that MATE proteins were found to be localized in the plasma membrane (80%), vacuoles (8.9%), chloroplast (4.4%), endoplasmic reticulum, nucleus (2.3%), cytoplasm and golgi body (1%) (Table S3).

A total of 14 gene loci encoding ALMT transporters were also identified from lentil genome available on Knowpulse (<https://knowpulse.usask.ca/>). These ALMT transporters were named LcALMT1 to LcALMT14 according to their physical location on the chromosomes. The length and molecular weights of the LcALMT protein had a narrow range, while the isoelectric points showed a broader variation (Table S3). The protein length, molecular weight and isoelectric point (pI) of LcALMT varied from 329 to 517 amino acids, 36.31 to 57.87 kDa and 5.75 to 9.24, respectively. All the fourteen LcALMTs had 6 transmembrane helices and were located on the plasma membrane (Table S3).

Chromosomal location of LcMATE and LcALMT genes

Based on the physical location of these 90 LcMATE genes, they were mapped on 7 lentil chromosomes (Fig. 1A). These were unevenly dispersed on all the chromosomes. The greatest number of LcMATE genes (33) were found on chromosome 2, while the lowest number of genes (5) were found on chromosome 3. Chromosomes

(See figure on next page.)

Fig. 1 **A** Chromosomal distribution of identified multidrug and toxic compound extrusion (MATE) genes in lentil. **B** Chromosomal distribution of lentil aluminum-activated malate transporter (ALMT) genes. Above each bar, the chromosomal number is displayed. Using data from Phytozome, the relative length of a chromosome indicates its size. The distance between genes is given in centimorgan (cM). **C** Duplication analysis of lentil MATE genes. Genes that are tandemly, dispersed, proximally duplicated are represented by the corresponding colors: green, red and blue. Singletons are represented in black. **D** Collinearity analysis of lentil MATE genes with *Arabidopsis* and soybean. The chromosomes of *Arabidopsis thaliana* are represented by five green bars, those of lentils by seven red bars, and those of soybeans by twenty blue bars. The collinear pairs of MATE genes are represented by orange curved lines

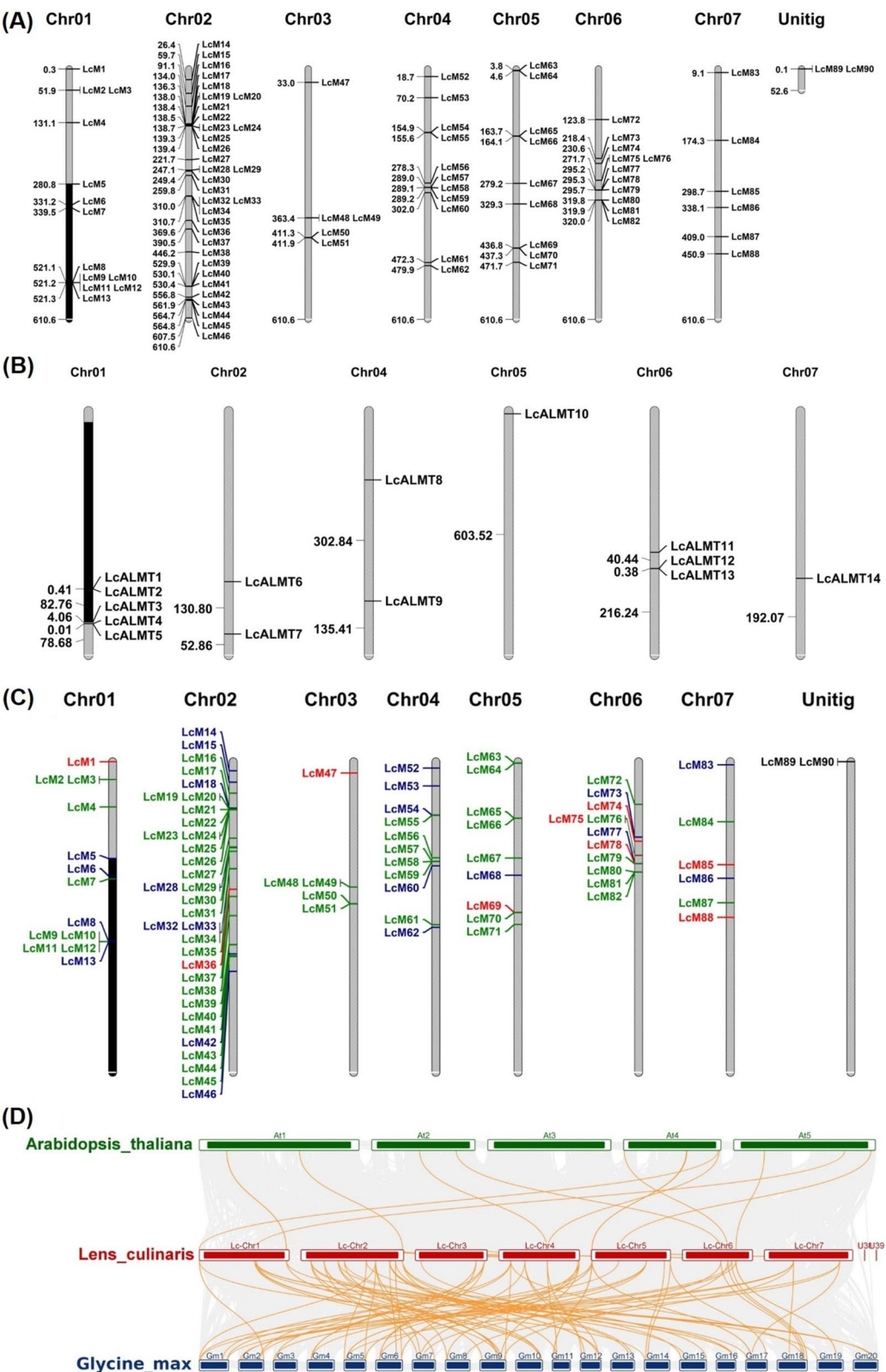


Fig. 1 (See legend on previous page.)

1, 5 and 7 had 13, 9 and 6 LcMATEs, respectively. Chromosome 4 and 6 had 11 LcMATEs each.

The 14 LcALMT genes were also scattered on 7 lentil chromosomes (Fig. 1B). Chromosome 1 contained the highest number of ALMT genes (5), whereas chromosomes 5 and 7 contained least (1 ALMT each). Chromosome 6 had 3 ALMTs, chromosomes 2 and 4 contained 2 ALMTs each, while chromosome 3 is devoid of any ALMT gene.

Gene duplication and collinearity analyses of LcMATE

According to the duplication studies, 88 of the 90 LcMATE genes (97.7%) were found to be involved in duplications. (Fig. 1C). We observed 57 (64.77%) LcMATE genes with tandem duplications distributed on all the chromosomes, 22 (25%) LcMATE genes with proximal duplications and 9 (10.23%) LcMATE genes were duplicated (Tables S4 and S5). This clearly revealed that tandem duplication have played a major role in amplification of MATE family in lentil.

Genome wide collinearity analysis of MATE gene families between lentil and other two dicot species i.e. *Glycine max* and *Arabidopsis thaliana* were conducted to gain a deeper comprehension of the evolutionary history of this gene family (Fig. 1D and Table S6). A total of 94 and 13 collinear gene pairs were recognised in *Glycine max* and *Arabidopsis thaliana*, respectively (Table S6). It was found that the syntenic genes of LcMATEs were found on all the chromosomes of *G. max* and 4 out of 5 chromosomes of *Arabidopsis*. There were more collinear blocks between lentil and soybean at the genome scale (Fig. 1D).

The Ka/Ks ratio aids in measuring molecular evolution and is a significant measure of selection pressure (Table S6). The majority of ortholog gene pairs had Ka/Ks values smaller than 1 (Table S6). The Ka/Ks ratio of 3 gene pairs was 1 indicating neutral selection pressure. It's interesting to note that both species have a number of collinear interactions, and that some collinear gene pairs were present in soybean and lentil but not in *Arabidopsis*. For example, 40 LcMATEs and 82 GmMATEs had collinear relationship (Table S6). These orthologous gene pairs had divergence times ranging from 26.37 to 217.20 million years ago (MYA).

Phylogenetic analysis of MATE and ALMT genes

The maximum likelihood (ML) approach was employed to build the phylogenetic tree and analyze the evolutionary

relationships of MATE genes (Fig. 2A). The 33 functionally characterized MATE protein sequences from various plant species were utilized in classifying the 90 LcMATE genes into 4 subfamilies. Among the subfamilies, subfamily S1 is the largest which contained 41 proteins followed by S2 (37), S3 (22), and S4 (23) subfamilies. The S1 subfamily consisted of three subgroups, S1-1, S1-2 and S1-3. S1-1 containing 14 sequences, including 10 LcMATE proteins and 4 known MATE transporters. S1-2 possessed 12 sequences, comprising 5 LcMATE proteins and 7 functionally characterized MATE transporters. There were 15 lentil MATE transporters grouped under S1-3 subgroup, which was devoid of any previously known protein. S2 subfamily had three subgroups, S2-1, S2-2 and S2-3. S2-1 each of which possessed 12 LcMATEs only. There were 3 LcMATE proteins as well as a known protein AtEDS5 (*Arabidopsis thaliana* enhanced disease susceptibility 5) in S2-2 subgroup. The S2-3 contained 5 LcMATEs protein sequences which were grouped together with 16 functionally characterized MATEs like GmFRD3b (*Glycine max* ferric reductase defective 3b), SbMATE (*Sorghum bicolor* MATE), BoMATE (*Brassica oleracea* MATE), etc. Subgroup 3 had 2 subgroups S3-1 and S3-2. The S3-1 had 7 MATE transporter proteins including 6 LcMATEs and AtALF5 (*A. thaliana* aberrant lateral root formation 5). The S3-2 possessed 13 LcMATE protein sequences and 2 characterized proteins (AtADS1 and AtZF14). Subgroup 4 had 2 subgroups S4-1 and S4-2. S4-1 had 9 LcMATEs together with two known proteins sequences viz. AtDTX1 (*A. thaliana* detoxification 1) and NtJAT1 (*N. tabacum* jasmonate-inducible alkaloid transporter 1). The S4-2 contained 12 LcMATEs grouped together.

The phylogenetic tree of ALMT was composed of 5 clades C1, C2, C3, C4 and C5. C1 was further subdivided into two subclades, C1-1 which contained 3 LcALMT, 3 OsALMT (*Oryza sativa* ALMT) and 5 AtALMT (*Arabidopsis thaliana* ALMT). C1-2 possessed 5 LcALMT, 1 OsALMT, 4 AtALMT and 1 GmALMT (*Glycine max* ALMT). C2 contained two LcALMT, three OsALMT and one AtALMT. C3 contained single LcALMT7 and C4 possessed LcALMT8 grouped with 6 previously characterized ALMTs involved in Al detoxification. Two LcALMT and AtALMT8 were clubbed as C5-1 while OsALMT1, HIALMT1, TaALMT1 and ScALMT were grouped under C5-2 (Fig. 2B).

(See figure on next page.)

Fig. 2 Phylogenetic tree of multidrug and toxic compound extrusion (MATE) and aluminum-activated malate transporter (ALMT) proteins.

A Phylogenetic association between 33 previously characterized MATE proteins and 90 LcMATE. Various colors represent four separate groups into which the MATE proteins were divided. **B** Phylogenetic association between 29 previously described ALMT proteins and 14 LcALMT. Various colors represent five distinct groups into which ALMT proteins were divided. MEGA X was used to create the phylogenetic trees using the Maximum Likelihood technique. Red bold dots were used to indicate the ones used for expressional quantification

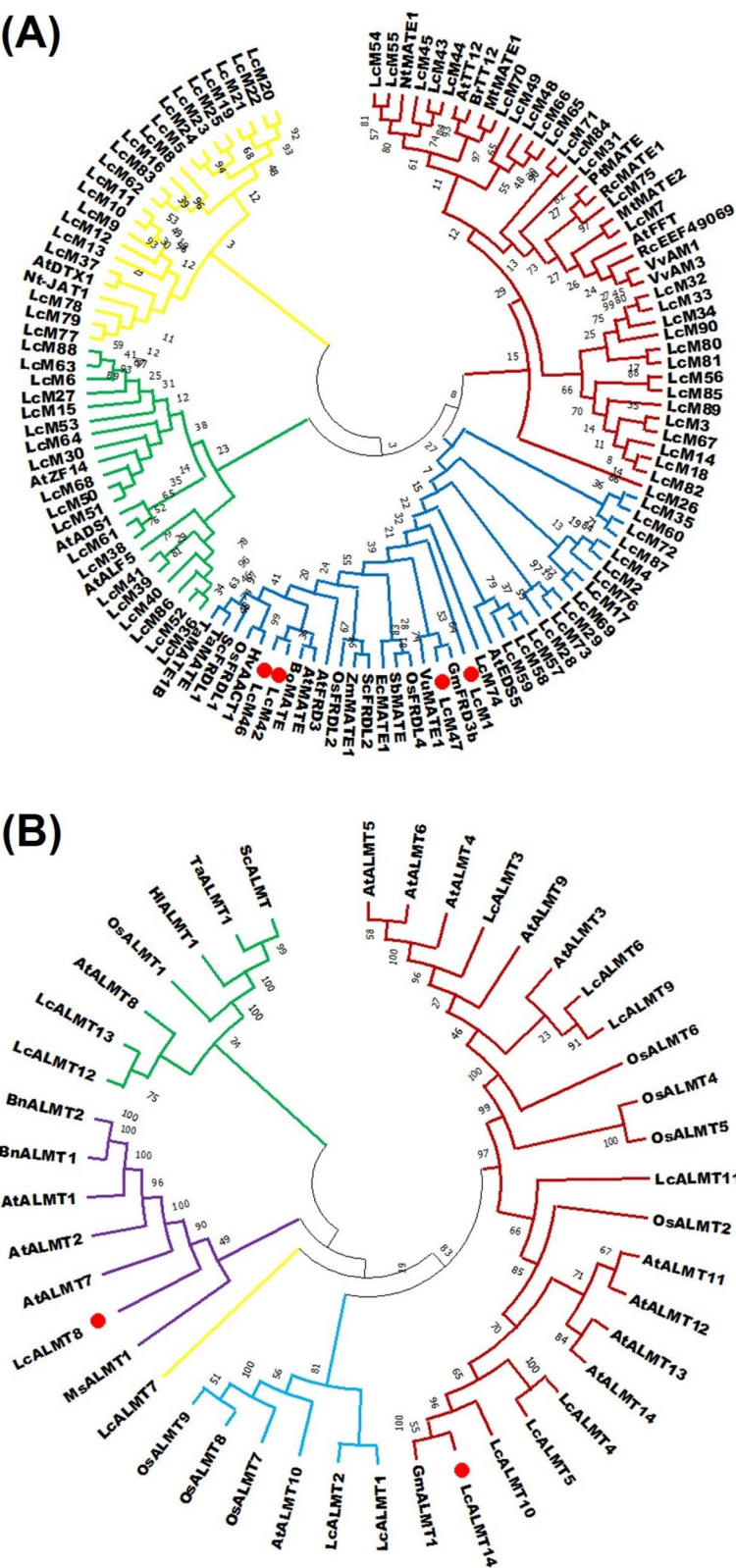


Fig. 2 (See legend on previous page.)

Fig. 3 Gene structures and protein motifs of lentil multidrug and toxic compound extrusion (MATE) and aluminum-activated malate transporter (ALMT). **A** Exon-Intron structure in lentil MATE as predicted by GSDS. The CDS and the UTR regions are represented by the pink and blue boxes, respectively. The black line are the introns. **B** Motifs of lentil MATE proteins as identified by MEME. Each colored box is a representation of a protein motif that MEME has identified. The box is positioned within the protein at the proper location. The order of MATE genes and proteins is according to the phylogenetic tree. **C** Exon-Intron framework in lentil ALMT. Magenta and green box exhibit the CDS and the UTR, respectively. The black lines are the introns. **D** Conserved motifs of lentil ALMT proteins. Each colored box depicts a protein motif that MEME identified. CDS (coding stand); GSDS: (gene structure display server) MEME (multiple EM for motif elicitation); UTR (untranslated region)

There is similarity in the types and sequences of motifs among the 14 LcALMT. The motifs in LcALMT varied in length from 27 to 50 amino acids. All of the identified LcALMTs were closely associated with Al-activated malate transport and carried the ALMT domain (Pfam No. PF11744) (Fig. 3D).

Cis-acting regulatory element analysis of LcMATE genes

The cis-acting regulatory elements (CREs) in the LcMATE genes' promoter area were found using the PlantCARE. Based on their functional annotation, these components can be categorized into three primary types: phytohormone responsive, light responsive, and stress responsive (Fig. 4A and Table S7). A total of 2792 CREs were discovered on LcMATE promoter regions, among which cis regulatory elements involved in stress responses were more enriched (15 for drought inducibility, 39 for defense and stress response, 115 for stress response, 48 for low temperature response and 508 for anaerobic response). Moreover, 795 cis regulatory elements (CREs) were found to be associated with phytohormones including auxin, abscisic acid, gibberellic acid, methyl jasmonic acid, salicylic acid and ethylene. Four members of LcMATEs (LcM1, LcM42, LcM46 and LcM47) grouped in S2-3 subgroup contained GGNVS element in their promoter region which can bind the Al resistance transcription factor 1 (ART1) suggesting their involvement in aluminium stress response (Fig. 4B).

Protein structure analysis and molecular docking of the citrate binding sites of LcMATE proteins

Protein sequences were aligned using multiple sequence alignment of the five deduced LcMATE grouped in subgroup 3 of subfamily 2 and representative MATE transporters known to be involved in aluminum detoxification via organic acid exudation in other plant species viz., *Arabidopsis*, rice bean, soyabean, cabbage and sorghum.

The citrate exuding motif (CEM) was extensively conserved and highlighted in Fig. 5A. To inspect the binding position of LcMATE to citrate, prediction of 3D structure and molecular docking were performed. In the present study, it was found that CEM was present in LcM1, LcM42, LcM46, LcM47 and LcM74 which contained 13, 15, 15, 18 and 17 citrate-binding sites, respectively (Fig. 5B). Remarkably, it was discovered that the amino acid residues identified in CEM Arg94, Ile97, Leu100, and Val101; Leu90, Phe91, Asn92, Ala94, and Ser95; Thr97; Phe160, and Val161 were involved in binding citrate in LcM42, LcM46, and LcM74, respectively. These sites were found in the central cavity of the LcMATE transporters, which is situated between the amino (N) and carboxy (C) terminals.

These LcMATEs shared 39.80–96.16% sequence identity amongst each other (Fig. S1A). The sequence analysis suggested that both LcM42 and LcM46 were highly similar to MATE-type citrate transporters *Brassica oleracea* MATE transporters (BoMATE) and *Arabidopsis thaliana* MATE (AtMATE) (Fig. S1A). LcM1 and LcM74 shared the lowest identity of 40–50% with known MATE-type citrate transporters. LcM47 was found to be 79.93% identical to GmFRD3b and 74.20% similar with VuMATE1 (Fig. S1A).

The transporters LcM42 and LcM46 had the same anticipated big intracellular loop between their 2nd and 3rd transmembrane domains, according to structure prediction using HMMTOP, and the amino acids found in the loop's first half were conserved. However, there was a difference in the predicted subcellular locations of LcM42 and LcM46 transporters. These transporters were localized to vacuole and plasma membrane, respectively. LcM47 had the extracellular loop between 2nd and 3rd transmembrane domain and was present on the plasma membrane. LcM1 and LcM74 lacked any of these characteristics found in typical citrate transporters (Fig. S1B).

Protein-protein interactions of LcMATE proteins

Protein-protein interactions were analysed using STRING software for five LcMATE transporters predicted to participate in Al detoxification (Fig. 6). LcM42 and LcM46 both were probable homologs of AtMATE1 and were predicted to interact with Lcu.2RBY.1g046260.1 (Fig. 6B and C). This protein was annotated as C2H2-type domain-containing protein and shared 81% sequence similarity with VuSTOP1 (*Vigna umbellata* sensitive to proton rhizotoxicity 1) (Fig. S2). LcM47 was predicted to interact with Lcu.2RBY.5g023960.1, Lcu.2RBY.2g097050.1, Lcu.2RBY.1g074220.1 and many more proteins (Fig. 6D). These were annotated as MYB family transcription factor, metal ion transporter NRAMP (natural resistance-associated macrophage protein) and aluminum-activated malate transporter 4-related respectively. LcM1 interacted with a single protein Lcu.2RBY.2g097350.1, annotated as adenine nucleotide alpha hydrolase-like domain kinase which showed interactions with other proteins (Fig. 6A). LcM74 formed a larger interactome and was found to interact with Lcu.2RBY.4g013480.1, Lcu.2RBY.2g041620.1 and Lcu.2RBY.2g017630.1 (Fig. 6E). All the three loci have been identified as members of MATE gene family in this study.

Expression patterns of MATE and ALMT genes

A total of 6 genes (LcM1, LcM42, LcM46, LcM47, LcALMT8 and LcALMT14) were chosen to determine the gene expression patterns in root by qRT-PCR analysis. These genes are homologous to already known Al-responsive genes. Under condition of Al stress, the expression of these genes was examined in roots

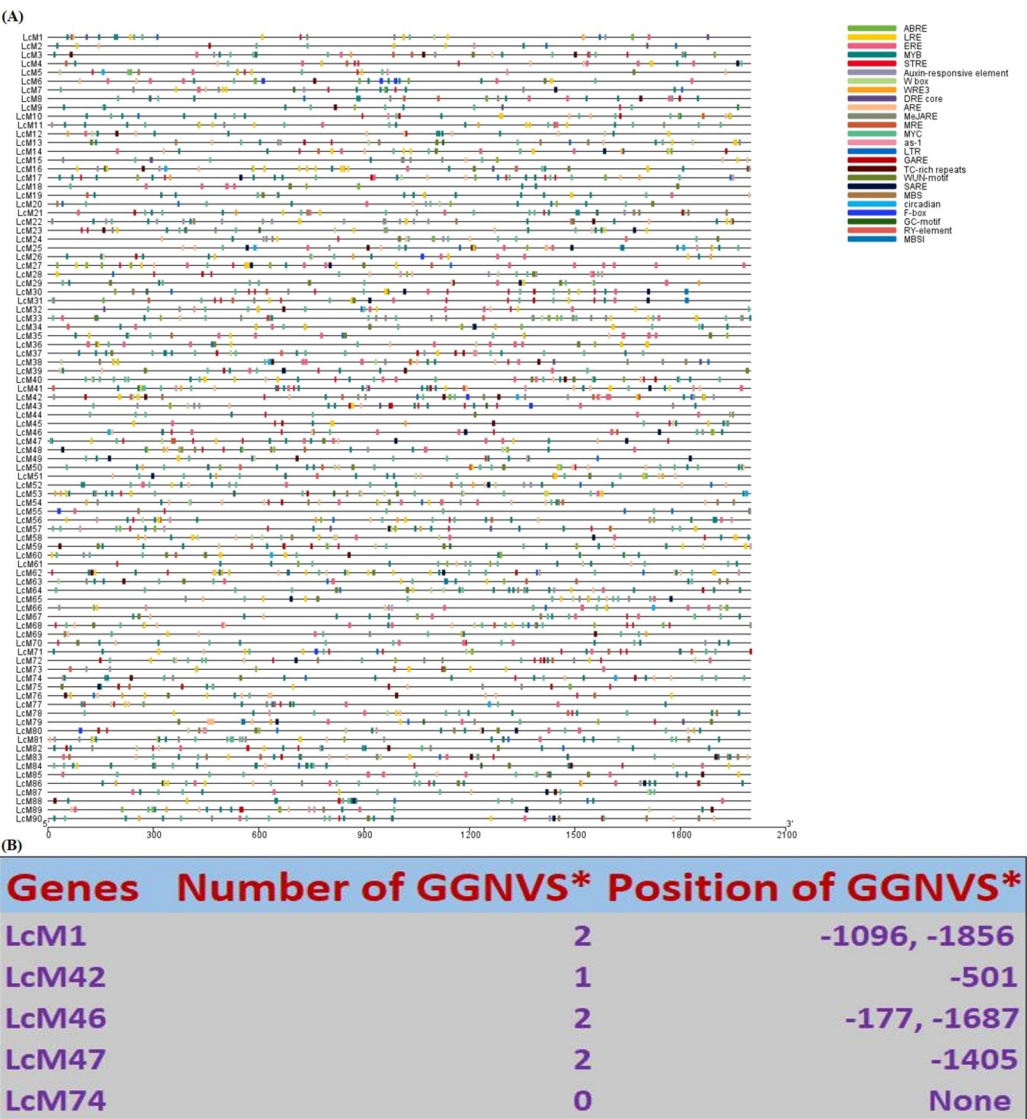


Fig. 4 Identification of cis-regulatory elements on LcMATE genes promoter. **A** Diagrammatic rendition of CREs (Cis-regulatory element) identified in LcMATE1 to LcMATE90. The line portrays the promoter region of the LcMATE genes. Coloured rectangular bars represented various CREs. The analysis for CREs was done in 2 kb upstream region of LcMATE genes. ABRE: Absciscic acid responsive element, MYB: Myoglobin, LRE: Light responsive element, ERE: Ethylene responsive element, STRE: Stress responsive element, Auxin responsive element, W box: WRKY binding box, DRE core: Drought responsive element, ARE: Anaerobic responsive element, MeJARE: Methyl Jasmonate responsive element, MRE: MYB binding site involved in light responsiveness, as-1: activation sequence-1, LTR: Low temperature responsive element, GARE: Gibberellic acid responsive element, WUN motif: Wound responsive motif, TC-rich repeats: cis-acting element involved in defense and stress responsiveness, SARE: Salicylic acid responsive element, MBS: MYB binding site involved in drought-inducibility, F-box: Floral organ protein related cis regulatory element, GC-motif: enhancer-like element involved in anoxic specific inducibility, RY element: maturation-specific gene promoters, MBSI: MYB binding site involved in flavonoid biosynthetic genes regulation. **B** Number and position of GGNVS element in the promoter region of five LcMATE genes grouped under S2-3 subfamily. The term GGNVS element is GGN(T/g/a/C)V(C/A/g)S(C/G), a cis-acting element of ART1 (Al Resistance Transcription Factor 1). The location is in relation to the initiation codon of each gene

(See figure on next page.)

Fig. 5 Multiple Sequence alignment (MSA), structural study and molecular docking models for citrate binding LcMATES classified under subgroup 3 of subfamily 2. **A** Amino acid sequence alignment was by multiple sequence alignment. The citrate exuding motif, which is the highlighted region, is a highly conserved first half of the loop between the second and third transmembrane that is 50 amino acids long. This loop is shared by the recognized citrate transporters and their likely lentil homologs. **B** Predicted structure of LcMATES using Phyre2, citrate binding positions in several MATE transporters, and citrate binding site LcM1, LcM42, LcM46, LcM47 and LcM74 transporters

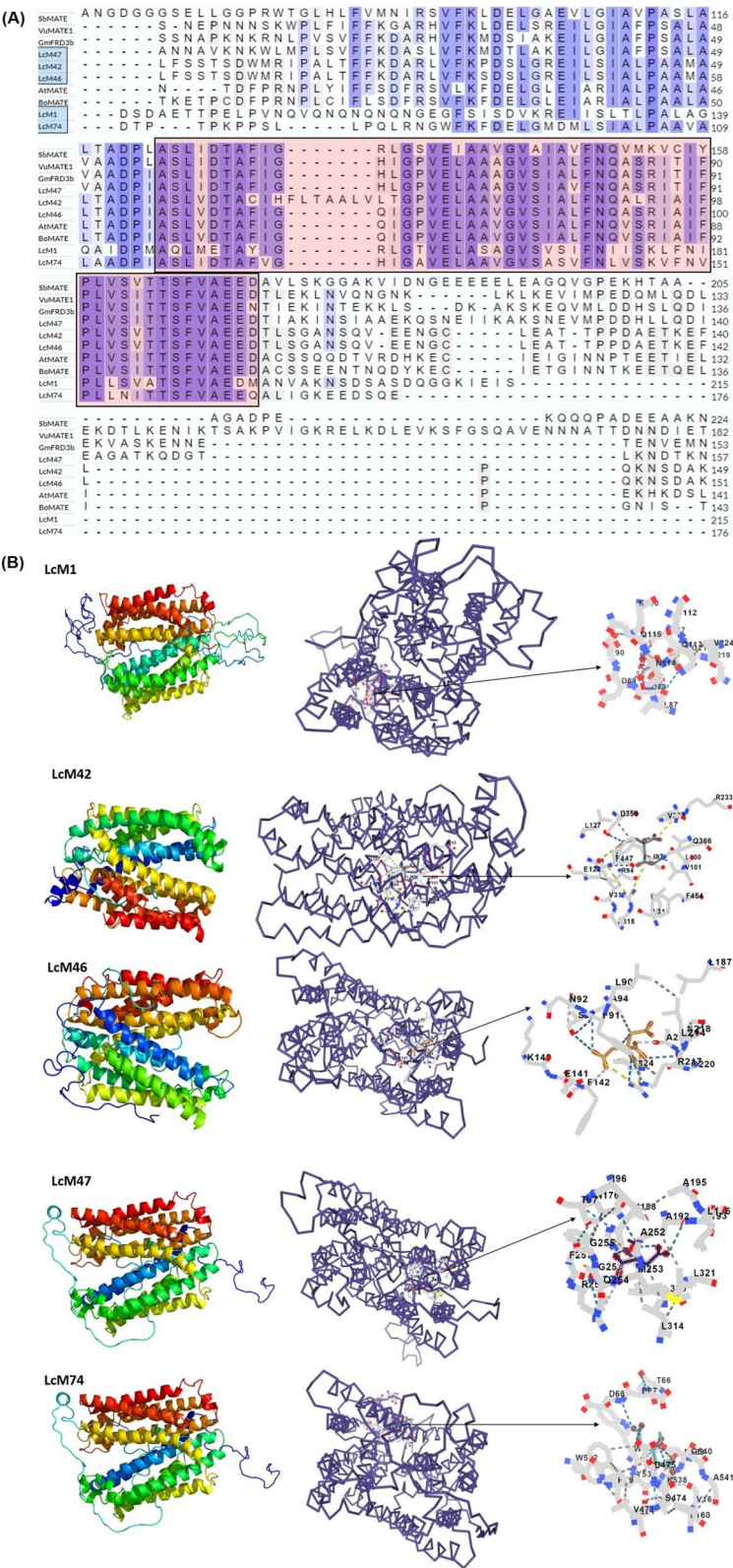
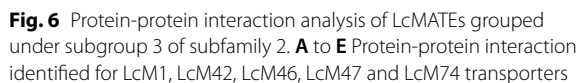


Fig. 5 (See legend on previous page.)



The multidrug and toxin extrusion (MATE) and aluminium activated malate transporter (ALMT) family members are localized on bio membranes of plant cells [56, 58]. These transporters play pivotal roles in the regulation of plant secondary metabolites and detoxification

The number of MATE in lentil (90) was larger than that reported in tomato (67) (Santos et al., [64], apple (66) [85], *Medicago truncatula* (70) [78], *Oryza sativa*

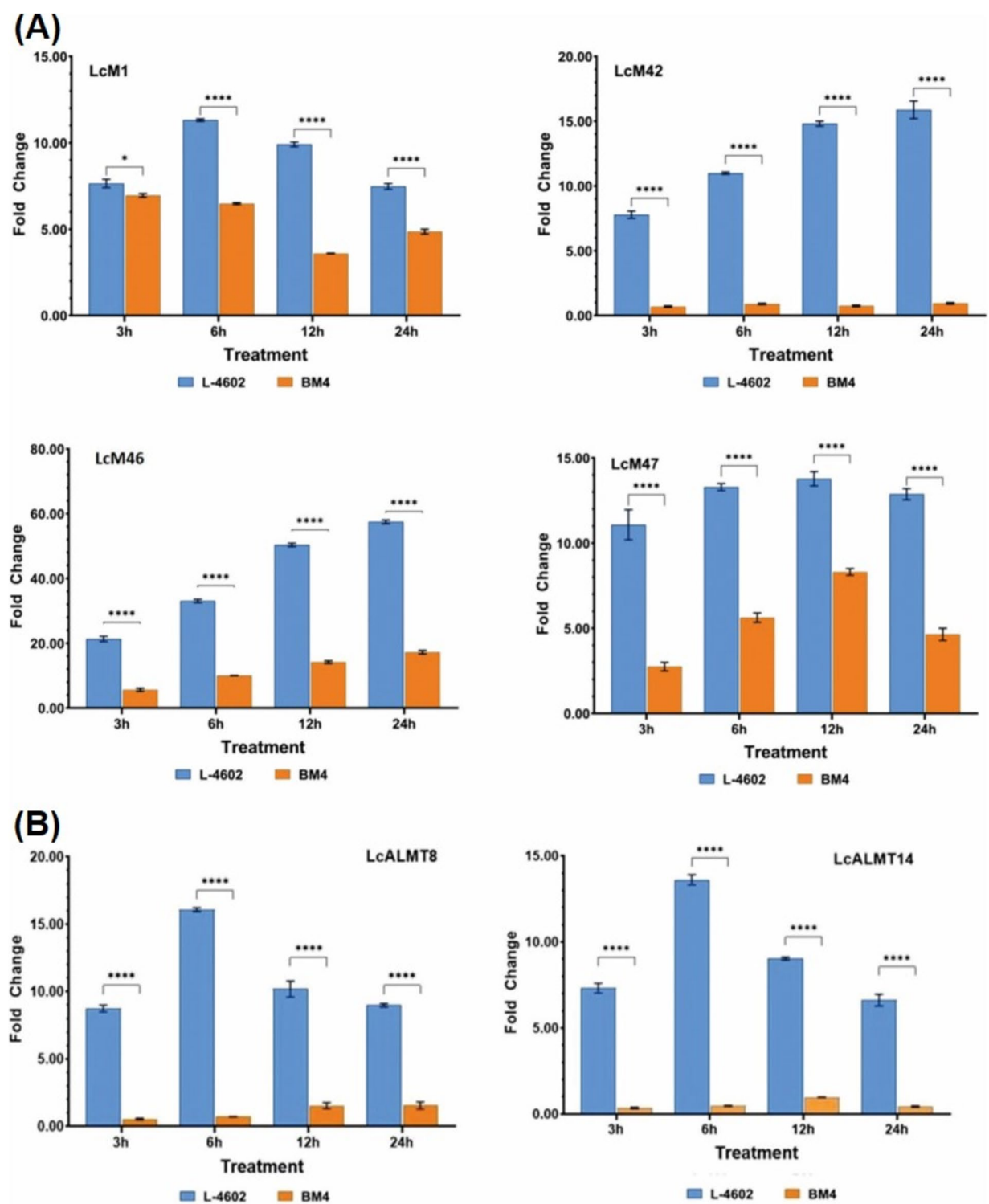


Fig. 7 Expression profiling of selected LcMATE and LcALMT genes in response to aluminum stress. **A** Relative expression of four LcMATE genes. **B** Relative expression of two LcALMT genes. Under aluminum stress, the expression patterns of these genes in roots were investigated. After treatment, the roots were taken out 0, 3, 6, 12, and 24 h later. Three biological repeats were averaged to provide each data point. Beta tubulin was used as the internal reference to standardize the transcript levels. As per Student's t test, **** $p < 0.0001$ and * $p < 0.05$

(53) [72] and *Arabidopsis thaliana* (56) [36]. Though the number was smaller than soybean MATE gene family (117) [40]. Vast number of MATE genes in lentil might be a repercussion of duplication events. It also signifies importance of the gene family in adaptation, survival, and evolutionary success by functional diversification, adaptive evolution, stress responsive versatility, and genomic plasticity. Gene duplication is one of the dominant driving forces in adaptive evolution of gene families and origin of new genes [53]. Gene duplication analysis in the present study showed that tandem duplication played major role in amplification of MATE family in lentil which was in accordance with previous studies in *Arabidopsis* and tomato [2, 64] (Fig. 1C, Table S4 and S5). So in case of lentil also, duplicated genes remained in close proximity on the same chromosome, which usually forms clusters of genes. Tandemly duplicated genes may initially exhibit redundancy, but over the time, they may go for neofunctionalization or subfunctionalization, acquiring new functions or specializing in subset of original roles.

Collinearity analysis and most ortholog gene pairs having Ka/Ks values smaller than 1 authenticated that LcMATE gene family was subjected to strong purifying selection during its evolutionary process (Table S6). Our results are close to those in other plants like potato and maize [3, 88], indicating that LcMATE genes are relatively conserved in the process of evolution and maintained functional stability. Lentil and soybean exhibited collinear relationship, indicating that LcMATE genes are closely linked to *Glycine max* during evolution (Fig. 1D, Table S6).

Phylogenetic analysis demonstrates evolutionary relatedness among groups of organisms, which can be used for the prediction of gene function and gene-gene associations [54]. In this study, phylogenetic analysis was performed with MATE and ALMT transporter protein families in lentil and other plant species including *Arabidopsis*, rice, wheat, rye, barley, jowar, tobacco, soybean, maize, *Medicago truncatula*, grape, cabbage, castor, eucalyptus, black cotton wood, and turnip. Analysis was conducted on 123 MATE proteins, 90 of which were predicted from the lentil genome. All these MATE proteins from various species were classified into four subfamilies, which was consistent with previous studies [40] (Fig. 2A). The functions of the LcMATE proteins can be predicted based on the known functionally characterized MATE transporters present on the same branch.

Among the known MATE proteins as shown in Fig. 2A, the members of subgroup 3 of subfamily 2 were clustered with AtFRD3 [59, 61], AtMATE [36, 41], OsFRDL1 [84], SbMATE [9], TaMATE1, 2 [20] and ZmMATE1 [50] and may be involved in secretion of citrate under Al stress. Subgroup S3-1 had lentil homologs of AtALF5 which is

known to function as efflux transporter for transport of secondary metabolites and helps in detoxifying harmful compounds by compartmentalizing them into vacuoles or transferring out of the cell [6].

A large difference in phylogenetic evolutionary classification of ALMT transporters exists among various plant species. The predicted ALMT genes in *Arabidopsis* (14), rice (9) and soybean (34) were grouped into three clusters [32], five clades [42] and two main groups having 6 subgroups [57], respectively. In lentil, 14 LcALMTs were grouped into 5 clades. In subclade 2 of clade 1, GmALMT1 known to be involved in malate exudation in response to low pH, aluminum toxicity and phosphorus deficiency was present [37]. The LcALMT14 was found in the same subclade indicating that this probably has similar function. Clade 4 possessed MsALMT1 [4], BnALMT1 and BnALMT2 [38], which are known to impart aluminum tolerance by malate exudation. The protein sequence LcALMT8 was present in the same clade and thus it might act as malate transporter for Al tolerance in lentil. Clade C5 had two subclades. Subclade C5-2 contained ScALMT1 [16] and TaALMT1 [65], which are known to play major role in aluminum tolerance by malate-Al complex exudation (Fig. 2B).

The proteins MsALMT1 and GmALMT1 have been characterized for aluminium detoxification [4, 37]. It was observed that LcALMT8 and LcALMT14 were closely related to MsALMT1 (50% sequence similarity) and GmALMT1 (65.85% sequence similarity), respectively (Fig. S3). These results showed that LcALMT8 and LcALMT14 genes can play important roles in aluminium detoxification.

Changes in the gene functions can be caused by changes in amino acid(s) by substitutions and/or variation in exon-intron structure [82]. The exon-intron framework and motif analysis of LcMATE and LcALMT revealed that genes within the same subfamily shared conserved motifs and had similar gene framework (Fig. 3). This implies that particular regulatory protein motifs and exon-introns may be linked to functional differentiation of various LcMATE subfamily genes, and the regulatory elements are preserved, leading to consistent expression patterns under specific conditions. The cis-regulatory elements play important role in gene transcriptional regulation occurring in response to various developmental changes and stress [80]. In this study, several cis-regulatory elements were identified (Fig. 4A and Table S7). GGNVS element [GGN(T/g/a/C) V(C/A/g) S(C/G)], the DNA-binding sequence of ART1 (Al resistance transcription factor 1) was involved in regulating the expression of 31 genes (including MATE), which confer Al tolerance in rice [73]. Based on this study, we found GGNVS element in 4 out of 5 LcMATEs were grouped

with other known MATEs involved in aluminum tolerance (Fig. 4B).

The structure and expressional profile of LcMATEs in subgroup 3 (S2-3) of subfamily 2, where most known MATE proteins were linked to iron translocation or aluminum detoxification, were further examined. A close relationship between LcM47, GmFRD3 (79.93% sequence similarity) and VuMATE (74.20% sequence similarity) was observed (Fig. S1A). More than 60% of the sequences of LcM42 and LcM46 were homologous to BoMATE and AtMATE (Fig. S1A). As expected for identified MATEs implicated in citrate transport, the potential citrate transporters LcM42 and LcM46 were discovered to have large intracellular loop between the second and third transmembrane domains and substantial sequence similarity in the first half of this loop (Fig. S1B) [41]. The MATE protein LcM46 was present in the plasma membrane, therefore citrate Al complex formed in root cells might be exuded in the rhizosphere, thereby alleviating Al toxicity. LcM42 was predicted to be localized on the tonoplast (Vacuolar membrane), and thus might be involved in sequestration of Al in the vacuoles, resulting in internal detoxification *via* organic acids similar to buckwheat [46] (Table S3). It is anticipated that LcM47 would be found in plasma membrane, which possesses a sizable extracellular loop connecting the second and third transmembrane domains (Fig. S1B).

Concrete evidences exist that the plants transport citrate by MATE transporters under Al stress [8, 71]. We found that most of the LcMATE proteins contained motif 6 and motif 9 indicating that these motifs were relatively conserved in the MATE family (Fig. 3B). The MATE proteins like SbMATE, VuMATE, AtMATE and ZmMATE involved in response against Al stress are known to have a highly conserved domain of citrate exuding motif (CEM) [76]. Amino acid residues identified in CEM were found to be involved in binding citrate so, under Al stress they may be crucial for citrate binding and delivery. Moreover, it was reported that lack of CEM in SbMATE (*Sorghum bicolor* MATE) resulted in loss of function [9]. Mutant of AtMATE exuded less citrate under Al stress [75]. Citrate transporting MATEs such as VuMATE and SbMATE mediate citrate flux into xylem, causing its further exudation into the rhizosphere to form Al-citrate complexes, thereby helps in mitigating Al toxicity [9, 43]. Meanwhile, MATE transporters like FeMATE2 in buckwheat transport the Al-citrate complexes formed in roots cells to the leaf cells, and deposit them in the vacuoles *via* golgi bodies [34]. In the present study, we found several citrate binding sites present in the CEM region of LcM42, LcM46, LcM47 and LcM74 (Fig. 5A and B), which could be crucial for citrate binding and transport under Al stress.

In this study, LcM42 and LcM46 both were predicted to interact with a plausible homolog of STOP1 in lentil (Lcu.2RBY.1g046260.1) (Fig. 6 and Fig. S2). STOP 1 may serve as regulatory hub that ensures MATE proeins are expressed under Al toxic conditions. This *in silico* analysis is in accordance with previous study where Al-resistance transcription factor STOP1 (C2H2-type zinc finger protein) played a major role in regulating MATE1 expression and thus its role in Al tolerance was confirmed [14, 87]. The interaction of the two genes with plausible homolog of STOP1 in lentil suggest that STOP1 could influence localization and stability of their corresponding proteins for coordinated response under stress condition.

Earlier reports indicated that MATE and ALMT family members were involved in secretion of organic acids in various plant species under aluminium stress condition [6, 17, 49, 60, 62]. According to Han et al. [24], expression pattern plays key role in determining the gene function. Our results showed differences in the expression patterns of LcMATE and LcALMT genes in roots of lentil at various time points i.e. 3, 6, 12 and 24 h under Al stress condition (Fig. 7A and B). The expression level of LcM46 was significantly higher as compared to other LcMATE genes (LcM1, LcM42, and LcM47) in Al-tolerant genotype (L-4602) than in sensitive genotype (BM-4) under all the time points in response to Al stress. Consequently, LcM46 might be a potential gene for tolerance against Al toxicity in lentil (Fig. 7A). Similarly, the dynamic expression profiling of both LcALMTs (LcALMT8 and LcALMT14) displayed a similar pattern of expression and reached the highest level at 6 h of Al stress in tolerant genotype under Al stress condition (Fig. 7B). The expression profiles of these MATE and ALMT genes predicted that they might be involved in Al detoxification.

Conclusions

This is the first comprehensive study of MATE and ALMT gene families and their expression analysis in response to aluminium stress in lentil (*Lens culinaris* Medikus). Evolutionary conservation of MATE and ALMT genes is suggested in lentil as they were not evenly distributed across chromosomes, and the primary mode of expansion was tandem duplication. A close association between lentil and soybean MATEs was found by collinearity analysis. Five LcMATE genes in subfamily 2 might play an important role in Al detoxification which showed similarity to citrate transporters already identified for Al tolerance. Interestingly, the sequences of LcMATE42 and LcMATE46 were highly comparable to those of previously published MATEs regulating Al tolerance and they were substantially expressed at the tips of the roots, under Al stress, suggesting lentil's ability to tolerate Al through citrate secretion. This work lays the groundwork for further investigation into the roles of LcMATE and

LcALMT genes and further understanding of the molecular mechanisms involved in aluminium stress. Future research can use gene editing technologies to examine the functional relevance of these potential genes and validate their involvement in Al detoxification. Additionally, the subcellular location of these proteins in the plasma membrane can be verified by confocal microscopy and GFP tagging. The protein-protein interactions can be proved by yeast two hybrid or dual luciferase systems.

Abbreviations

LcMATE	<i>Lens culinaris</i> Multidrug and toxic compound extrusion
LcALMT	<i>Lens culinaris</i> Aluminium activated malate transporter
Aa	Amino acids
CEM	Citrate Exuding Motif
KDa	Kilodalton
MW	Molecular Weight
Ka	Nonsynonymous Substitutions
Ks	Synonymous Substitutions
ABRE	Abscisic acid responsive element
ERE	Ethylene responsive element
ARE	Anaerobic responsive element
GARE	Gibberellic acid responsive element
SARE	Salicylic acid responsive element
W box	WRKY binding box
MeJARE	Methyl Jasmonate responsive element
MBS	MYB binding site involved in drought-inducibility
TC rich repeats	cis-acting element involved in defense and stress responsiveness
DRE	Drought responsive element
LTR	Low temperature responsive element
STRE	Stress responsive element
WUN motif	Wound responsive motif
MBSI	MYB binding site involved in flavonoid biosynthetic genes regulation
MYB	Myeloblastosis viral oncogene homolog
MYC	Myelocytomatosis viral oncogene homolog
OAs	Organic Acids
qRT PCR	Quantitative Real Time Polymerase Chain Reaction
T	Divergence time
MYA	Million years ago

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06086-9>.

Supplementary Material 1. Supplementary Fig. S1. Sequence similarity matrix and Structure prediction of LcMATEs. (A) Sequence similarity matrix of LcM1, LcM42, LcM46, LcM47 and LcM74 with other previously characterized MATEs involved in aluminum stress tolerance via citrate exudation. (B) Structure of LcMATE proteins as predicted by HMMTOP. Twelve transmembrane helices and a large intracellular loop was seen between 2nd and 3rd transmembrane domains in LcM42 and LcM46.

Supplementary Material 2. Supplementary Fig. S2. Sequence similarity matrix of VuSTOP1 with its probable homolog in lentil (Lcu.2RBY.1g046260.1).

Supplementary Material 3. Supplementary Fig. S3. Sequence similarity matrix of LcALMTs. (A). Sequence similarity matrix of LcALMT8 with known ALMTs involved in aluminum stress tolerance via malate exudation. (B). Sequence similarity matrix of LcALMT4, LcALMT5, LcALMT10 and LcALMT14 with GmALMT1 known to be involved in responses against phosphate starvation.

Supplementary Material 4. Supplementary File S1. Gene, coding and protein sequences of LcMATEs.

Supplementary Material 5. Supplementary File S2. Gene, coding and protein sequences of LcALMTs.

Supplementary Material 6. Supplementary File S3. The protein sequences of 33 known MATEs and 29 known ALMTs used for phylogenetic analysis.

Supplementary Material 7. Supplementary Table S1. List of previously reported MATE and ALMT transporters from other plant species.

Supplementary Material 8. Supplementary Table S2. List of forward and reverse primers used for qRT-PCR analysis.

Supplementary Material 9. Supplementary Table S3. Details of the gene loci of identified LcMATE and LcALMT proteins including their chromosomal location, isoelectric pH, molecular weight, number of amino acids, number of transmembrane helices and subcellular localization.

Supplementary Material 10. Supplementary Table S4. Gene duplications as predicted in LcMATE gene family as per MCScanX.

Supplementary Material 11. Supplementary Table S5. Tandem duplicated genes recognised in LcMATE gene family.

Supplementary Material 12. Supplementary Table S6. One-to-one orthologous relationships between MATE genes of *Lens culinaris* and other two plant species (*Arabidopsis thaliana* and *Glycine max*).

Supplementary Material 13. Supplementary Table S7. Cis-regulatory elements recognized in LcMATEs.

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Authors' contributions

DS and AT conceived and designed the experiments. AT, JB, DS, JA and AHA performed the bioinformatics analysis. DS, AT, JT, JA, AHA and MP drafted the manuscript. All authors read and approved the final manuscript.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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